

PROFESSIONAL SUMMARY

ML Engineer with 2+ years of production experience in recommendation systems, time-series forecasting, and privacy-preserving machine learning. Skilled in Python, PyTorch, and cloud-based MLOps. MSE in Computer Science at Stanford University.

EDUCATION

Master of Science — Computer Science Stanford University, Stanford, CA	Aug 2023 – May 2025 GPA: 3.82/4.0
Bachelor of Science in Computer Science University of Michigan, Ann Arbor, MI	Aug 2019 – May 2023 GPA: 3.91/4.0

PROFESSIONAL EXPERIENCE

ML Engineer — Prismatic Labs Jul 2023 – Present
Series A startup building AI-powered personalization infrastructure for media platforms

- **Built a contextual multi-armed bandit recommendation system** using LinUCB with user embedding lookups, increasing click-through rate by 22% and reducing cold-start latency from 800ms to under 200ms.
- **Trained a Temporal Fusion Transformer** for multi-horizon demand forecasting across 400+ product SKUs, achieving 9.3% MAPE vs. 14.1% for the previous ARIMA baseline in production.
- **Designed a feature store** using Feast + Redis for real-time feature serving at <10ms p99, decoupling training and inference pipelines and eliminating training-serving skew across 6 downstream models.
- **Instrumented model monitoring** with Evidently AI for data drift and concept drift detection, triggering automated retraining workflows that reduced mean-time-to-recovery from 3 days to under 4 hours.

PROJECTS

Federated Learning Framework for Privacy-Preserving Clinical NLP
Python, PyTorch, Flower, HuggingFace, Differential Privacy

- Implemented a federated fine-tuning system for clinical BERT across 8 simulated hospital nodes using the Flower framework, achieving within 1.4% F1 of centralized training with zero raw data leaving node boundaries.
- Applied per-sample gradient clipping and Gaussian noise injection (DP-SGD) with $\epsilon = 3.2$, and profiled the privacy-utility tradeoff across clipping thresholds using Rényi differential privacy accounting.
- Benchmarked FedAvg, FedProx, and SCAFFOLD aggregation strategies on non-IID clinical note splits, finding FedProx reduced client drift by 31% on the most heterogeneous partition.

Multi-Fidelity Bayesian Hyperparameter Optimization Pipeline
Python, Optuna, Ray Tune, Weights & Biases, BOHB

- Built a hyperparameter search pipeline using BOHB (Bayesian Optimization + HyperBand) that cut total GPU-hours for architecture search by 58% compared to random search across image classification benchmarks.
- Implemented a Gaussian Process surrogate with Matérn kernel and expected-improvement acquisition, integrating early stopping via successive halving to eliminate underperforming trials after 2 epochs.
- Logged all trials to Weights & Biases with parallel-coordinates sweep visualization; results reproduced across 5 seeds with <0.3% variance in final validation accuracy.

Graph Neural Network for Drug-Target Interaction Prediction
Python, PyTorch Geometric, DGL, RDKit, DeepChem

- Modeled drug-target interaction as a bipartite link-prediction task on a heterogeneous graph of 12K compounds and 4K protein targets, using message-passing GNN with edge-type-specific weights.
- Achieved AUROC of 0.913 on the BindingDB test split, outperforming DeepDTA CNN baseline by 6.2 points; ablated attention-pooling vs. mean-pooling on molecular subgraphs to explain the gain.

PUBLICATIONS

Federated Fine-Tuning of Clinical Language Models under Differential Privacy Constraints	2024
Multi-Fidelity Bayesian Optimization for Neural Architecture Search on Imbalanced Tabular Data	2023

SKILLS

Machine Learning & AI: Recommendation Systems, Time-Series Forecasting, Federated Learning, GNNs, Transformers, RL, NLP
Frameworks & Tools: Python, PyTorch, JAX, Scikit-Learn, HuggingFace, Optuna, Ray Tune, ONNX, Git
MLOps & Infrastructure: AWS (SageMaker, S3, EKS), Feast, MLflow, Weights & Biases, Docker, Kubernetes
Databases: PostgreSQL, Redis, Pinecone, MySQL